Name:

AP Calculus BC Test #3 Review

1. Let $f(x) = e^{x^3} + sin(x^3)$
   a) Write the first four nonzero terms and the general term of the Maclaurin series for $sin\ x$ and write the first four nonzero terms of the Maclaurin series for $sin\ (x^3)$.
   b) Write the first four nonzero terms and the general term of the Maclaurin series for e^x and write the first four nonzero terms of the Maclaurin series for e^{x^3}.
   c) Use part (a) and (b) to write the first four nonzero terms of the Maclaurin series for f.

a) $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{7!} + \ldots + (-1)^n \frac{x^{2n+1}}{(2n+1)!}$

$\sin(x^3) = x^3 - \frac{(x^3)^3}{6} + \frac{(x^3)^5}{120} - \frac{(x^3)^7}{7!} + \ldots + (-1)^n \frac{(x^3)^{2n+1}}{(2n+1)!}$

$\sin(x^3) = x^3 - \frac{x^9}{6} + \frac{x^{15}}{120} - \frac{x^{21}}{7!} + \ldots + (-1)^n \frac{x^{6n+3}}{(2n+1)!}$

b) $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \ldots + \frac{x^n}{n!}$

$e^{x^3} = 1 + x^3 + \frac{(x^3)^2}{2} + \frac{(x^3)^3}{6} + \ldots + \frac{(x^3)^n}{n!}$

$e^{x^3} = 1 + x^3 + \frac{x^6}{2} + \frac{x^9}{6} + \ldots + \frac{x^{3n}}{n!}$

c) $\boxed{1 + 2x^3 + \frac{x^6}{2} + \frac{x^{12}}{24}}$

2. Let $f(x) = \frac{1}{1+5x}$

a) Write the first four terms and the general term of the Maclaurin series for f(x).

b) What is the radius and the interval of convergence for the series in part (a).

c) Let g be a function given by $g(x) = \int_0^x f(t)dt$, find the first for terms and the general term of the Maclaurin series for $g(x)$.

a) $1-(5x)+(5x)^2-(5x)^3+\ldots+(-1)^n(5x)^n$

$1-5x+25x^2-125x^3+\ldots+(-1)^n(5x)^n$

b)

RTAC for $\sum_{n=0}^{\infty}(-1)^n(5x)^n$

$\lim_{n\to\infty}\left|\frac{(5x)^{n+1}}{(5x)^n}\right| \to \lim_{n\to\infty}\left|\frac{\cancel{(5x)^n}\cdot 5x}{\cancel{(5x)^n}}\right| \to \lim_{n\to\infty}|5x| = |5x|<1$

Center: $x=0$ $\quad -1<5x<1$ $\quad$ ROC: $\frac{1}{5}$ $\quad$ IOC: $(-\frac{1}{5},\frac{1}{5})$

$-\frac{1}{5}<x<\frac{1}{5}$

when $x=-\frac{1}{5}$

$\sum_{n=0}^{\infty}(-1)^n(5(-\frac{1}{5}))^n$

$\sum_{n=0}^{\infty}(-1)^n(-1)^n$

$\sum_{n=0}^{\infty}(-1)^{2n}$

$\sum_{n=0}^{\infty}(1)^n$

geometric, $r=1$

$|r|<1$ $\quad |1|=1$

The geometric series diverges

$x=\frac{1}{5}$

$\sum_{n=0}^{\infty}(-1)^n(5(\frac{1}{5}))^n$

$\sum_{n=0}^{\infty}(-1)^n(1)^n$

$\sum_{n=0}^{\infty}(-1)^n$

geometric, $r=-1$

$|r|<1$ $\quad |-1|=1$

The geometric series diverges

c) $g(x)=\int_0^x f(t)\,dt$

$\hookrightarrow$ t dummy variable

$g(x)=\int_0^x(1-5t+25t^2-125t^3+\ldots+(-1)^n(5t)^n)\,dt$

$$\boxed{g(x)=x-\frac{5}{2}x^2+\frac{25}{3}x^3-\frac{125}{4}x^4+\ldots+\frac{(-1)^n(5)^n x^{n+1}}{n+1}}$$

3.

a) Find the first four nonzero terms and the general term of the Maclaurin Series generated by $f(x) = \frac{x^3}{1+x}$

b) Find the interval of convergence of the series in part (a). ROC & center

c) Write the fifth degree Maclaurin Polynomial for f', the derivative of f at x=0.

a) $x^3(1 - x + x^2 - x^3 + \ldots + (-1)^n x^n)$

$\hookrightarrow x^3 - x^4 + x^5 - x^6 + \ldots + (-1)^n x^{n+3}$

b) RTAC for $\sum_{n=0}^{\infty} (-1)^n x^{n+3}$

$\lim_{n\to\infty} \left| \frac{x^{n+4}}{x^{n+3}} \right| \to \lim_{n\to\infty} \left| \frac{\cancel{x^n} \cdot x^4}{\cancel{x^n} \cdot x^3} \right| \to \lim_{n\to\infty} |x| = |x| < 1 \quad -1 < x < 1$

ROC: 1, center: $x=0$

IOC: $(-1, 1)$

when $x = -1$

$\sum_{n=0}^{\infty} (-1)^n (-1)^{n+3}$

$\sum_{n=0}^{\infty} (-1)^{2n+3}$

geometric, $r=-1$

$|r|<1$, $|-1|=1$

the geo series diverges

$x = 1$

$\sum_{n=0}^{\infty} (-1)^n (1)^{n+3}$

$\sum_{n=0}^{\infty} (-1)^n$

geometric, $r=-1$

$|r|<1$, $|-1|=1$

the geo series diverges

c) $f(x) = x^3 - x^4 + x^5 - x^6$

$\boxed{f'(x) = 3x^2 - 4x^3 + 5x^4 - 6x^5}$

a) Find the first four nonzero terms in the Taylor Series generated by $f(x) = \sqrt{1+x}$ at $x = 0$.

b) Use the result found in part (a) to find the first four nonzero terms in the Taylor Series for $g(x) = \sqrt{1+x^2}$ at $x = 0$.

c) Find the first four nonzero terms for the Taylor Series at $x = 0$ for the function such that $h'(x) = \sqrt{1+x^2}$ and $h(0) = 5$.

a) $f(x) = \sqrt{1+x}$ $\quad f(0) = 1$

$f'(x) = \frac{1}{2}(x+1)^{-1/2}$ $\quad f'(0) = \frac{1}{2}$

$f''(x) = -\frac{1}{4}(x+1)^{-3/2}$ $\quad f''(0) = \frac{-1}{4}$

$f'''(x) = \frac{3}{8}(x+1)^{-5/2}$ $\quad f'''(0) = \frac{3}{8}$

$P(x) = 1 + \frac{1}{2}x - \frac{\frac{1}{4}}{2!}x^2 + \frac{\frac{3}{8}}{3!}x^3$

$\boxed{P(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3}$

b) $1 + \frac{1}{2}x^2 - \frac{1}{8}(x^2)^2 + \frac{1}{16}(x^2)^3$

$\boxed{1 + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \frac{1}{16}x^6}$

c) $\int h'(x) = h(0) + \int_0^x \left(1 + \frac{1}{2}t^2 - \frac{1}{8}t^4 + \frac{1}{16}t^6\right) dt$

$\hookrightarrow \boxed{5 + x + \frac{1}{6}x^3 - \frac{1}{40}x^5}$